

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:20

Annexure A
MCQ TEST

Criteria	Conceptual Understanding(3)	Accuracy of Answer(2.5)	Application of Knowledge(2)	Completeness of Response(1.5)	Clarity & Presentation(1)	Total(10)
Weightage	30%	25%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I A. LOKESH KUMAR(192524157) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A. LOKESH KUMAR

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Wayground

Summary - Quizizz

Ask Gemini

wayground.com/join/game/U2FsdGVhX19VcuApwd3pqRpS6mEc8POO%252BJl78mhVb%252Bh4hGml39hKXjBwEJAwxZVyat5ZGhuwq06HitlqkyXg%253D%253D?page=nps&page=...

DWDM UNIT 1 - QUIZ - CSA1607



Congratulations you finished the quiz, LOKESH.

23³⁵

18200

+1020

Rank

Score

Coins

Play Again

Find New Quiz

Shop

Earn upto 1000 coins

SUMMER DROP

Shop now >



Play mini games

Play now

Accuracy

60%

Click here to refresh rank

Performance Stats

50 questions

30 Correct

20 Incorrect

20 s time/question

6 Streak

Review Questions

Click on the questions to see answers

Study Flashcards

1. Which SQL command is used to retrieve data from a

CO2- Assessment Tool-1

1. Suppose the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

- (a) Calculate Mean and Median of the data?

$$\text{Mean}(\bar{x}) = \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{809}{27}$$

$$\text{Median: } \frac{n+1}{2} = \frac{27+1}{2} = 14^{\text{th}} \text{ value}$$

Median : 25

- (b) What is Mode?

$\therefore$ 25 and 35 are repeating four times since it is a bimodal.

- (c) What is Midrange?

$$\text{Midrange: } \frac{\text{maximum} + \text{Minimum}}{2} = 41.5$$

$$\therefore \text{Midrange: } 41.5$$

6) Can you find the first quartile (Q_1)
and third quartile (Q_3)

$Q_1 = 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 23, 25, 25$

Middle Value = 20

$Q_3 = 20$

Upper half = 20, 23, 33, 33, 35, 35, 35, 35, 40, 45, 46, 54, 70

Middle Value = 35

$Q_3 = 35$

= 3

7)

IQR =

$Q_3 - Q_1$

$35 - 20$

~~15~~ 15

8) Five number Summary

~~Minimum~~

Statistic

Value

Minimum

13

Q_1

20

Median

25

Q_3

35

Maximum

70

(P) Box Plot:



13 15 16 19 20 21 22 25 30 33 35 36 40 45 46 52 60

Q)

Given the following measurements for the Variable Age :

18, 22, 25, 42, 28, 43, 33, 35, 36, 29 . Standardize the Variable by the following.

(a) Compute the mean absolute deviation of age.

Age = 18, 22, 25, 42, 28, 43, 33, 35, 36, 29

no of Observation
n=10

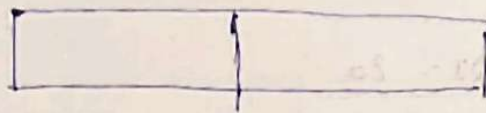
Mean: $\bar{x} = \frac{\sum x_i}{n}$

$$\frac{18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 36 + 29}{10}$$

$$\frac{380}{10} = 38$$

Mean = 38

(P) Boxplot:



13 15 16 19 20 21 22 25 30 33 35 36 40 45 46 52

(Q)

Given the following measurements for the Variable Age:

18, 22, 25, 42, 28, 43, 33, 35, 56, 29. Standardize the Variable by the following.

(a) Compute the mean absolute deviation of age.

Age = 18, 22, 28, 42, 29, 43, 33, 35, 56, 29

no of Observation
n=10

$$\text{Mean} = \frac{\sum_{i=1}^n x_i}{n}$$

$$= \frac{18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 29}{10}$$

$$= \frac{330}{10} = 33$$

Mean = 33

(X) Age	$ x - \bar{x} $	
18	-15	
22	-11	
25	-8	
42	9	
28	5	
45	10	
33	6	
35	2	
56	73	
28	5	
Σ	$\Sigma = 28$	

$$\Sigma |x - \bar{x}| = 88$$

- ③ Suppose that the data for analysis includes the attribute age. The age values are for data tuples are (in increasing order) 18, 19, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 33, 35, 35, 35, 36, 40, 43, 46, 52, 70. Use smoothing by bin means to smooth the above data. Using bin depth of 3 illustrate for steps.

Sol) Total no. of bins: $\frac{\text{Total}}{\text{bin}} = \frac{27}{3} = 9$

$$\frac{(x)_{\text{min}} - (x)_{\text{max}}}{(x)_{\text{min}} - (x)_{\text{max}}}$$

B _{in}	Original Values	Mean	Smoothed Value
B ₁	13, 15, 16	14.67	14.67, 14.67, 14.67
B ₂	16, 19, 20	18.33	18.33, 18.33, 18.33
B ₃	20, 21, 22	21	21, 21, 21
B ₄	22, 25, 25	24	24, 24, 24
B ₅	25, 25, 30	26.67	26.67, 26.67, 26.67
B ₆	33, 33, 35	33.67	33.67, 33.67, 33.67
B ₇	35, 35, 35	35	35, 35, 35
B ₈	36, 40, 45	40.33	40.33, 40.33, 40.33
B ₉	46, 52, 70	56	56, 56, 56

4. Suppose a group of
Use the two methods below to
normalize the following group of data:

200, 300, 400, 600, 1000

(a) min-max normalization by setting min=0 and max=1

(b) Z-score normalization

501.

(a)

$$X' = \frac{X - \min(X)}{\max(X) - \min(X)}$$

Minimum : 200

Maximum : 1000

max - min : 800

Original Value (x)	Calculation	Normalized Value (x')
200	$(200 - 200) / 800 = 0/800$	0.00
300	$(300 - 200) / 800 = 100/800$	0.125
400	$(400 - 200) / 800 = 200/800$	0.250
600	$(600 - 200) / 800 = 400/800$	0.500
1000	$(1000 - 200) / 800 = 800/800$	1.000

(b)

$$Z = \frac{X - \mu}{\sigma}$$

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$$

Mean = 500

X	X - μ	(X - μ) ²
200	-300	90,000
300	-200	40,000
400	-100	10,000
600	100	10,000
1000	500	2,50,000
Total		4,00,000

$$\sigma^2 = \frac{4,00,000}{5} = 80,000$$

$$\sigma = \sqrt{80,000}$$

$$= 282.84$$

Population Standard deviation = 282.84

(W) χ^2 Calculation Z -Score

200 $(200-500)/282.84 = -1.06$

300 $(300-500)/282.84 = -0.71$

400 $(400-500)/282.84 = -0.35$

600 $(600-500)/282.84 = 0.35$

1000 $(1000-500)/282.84 = 1.77$

$$\chi^2 = \frac{\sum (O - E)^2}{E} = \frac{200 + 900 + 1000 + 1000 + 1000}{2} = 2000$$

X	$X - \bar{X}$	$(X - \bar{X})^2$
200	-300	90000
300	-200	40000
400	-100	10000
600	100	10000
1000	500	250000
Total		390000

$$\sigma^2 = \frac{\sum (X - \bar{X})^2}{N} = \frac{390000}{2000} = 195$$

$$\sigma = \sqrt{195} = 13.96$$

Standard deviation = 13.96

5

Suppose a group of 12 Sales Price records has been Sorted as follows:

5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. Partition them into three bins by each of the following methods.

Sol.)

(a) Equal frequency Partitioning (b) Equal frequency Partitioning

Since there are 12 records and 3 bins.

$$\frac{12}{3} = 4.$$

Each bin contains 4 values.

Bin	Values
Bin 1	5, 10, 11, 13
Bin 2	15, 35, 50, 55
Bin 3	72, 92, 204, 215

(b) Equal width Partitioning.

Step 1: Find width

Formula:

Width =

$$\frac{\text{Maximum} - \text{Minimum}}{\text{Number of Bins}}$$

$$= \frac{215 - 5}{3} = \frac{210}{3} = 70$$

Step 2: Create Bins

Bin	Range	Values
Bin 1	5-74	5, 10, 11, 13, 15, 35, 50, 55, 72
Bin 2	75-144	92
Bin 3	145-215	204, 215

(C) Clustering

Clustering groups similar values together.

A reasonable partition is:

Cluster	Values
Cluster 1	5, 10, 11, 13, 15
Cluster 2	35, 50, 55, 72, 92
Cluster 3	204, 215

⑥ Suppose a hospital tested the age and body fat data for 10 randomly selected adults with the following result:

Age	23	23	27	27	39	41	47	49	50
% fat	9.5	16.5	7.8	17.8	21.4	28.9	27.4	27.2	31.2
Age	52	54	54	56	57	58	58	60	61
% fat	24.6	42.5	28.8	23.4	30.2	34.1	32.9	41.2	33.7

(a)

 $n = 12$

Mean

$$\bar{x} = \frac{\sum x}{n}$$

$$F \quad \begin{array}{r} 826 \\ \hline 18 \end{array}$$

2 46,44

Mean: 46.44

Median : $\frac{n}{2}$ - g^{th} or 10^{th} value.

$$50 + 52 = 102$$

Medion

Median = 51

standard deviation:

Age (x)	$x - \bar{x}$	$(x - \bar{x})^2$
23	-23.44	549.07
23	-23.44	549.07
27	-19.44	378.01
27	-19.44	378.01
39	-7.44	55.35
41	-5.44	29.60
47	0.56	0.31
49	2.56	6.55
50	3.56	12.67
52	5.56	30.92
54	7.56	57.16
54	7.56	57.16
56	9.56	91.39
57		
58		
58		
60		
61		

(C) Scatter Plot

or Fat

45

40

35

30

25

15

10

20

30

40

50

60

Age

Age

Board

Median (Q2)

$$= \frac{50+51}{2} = 50.5$$

Q1 = 39

Third Quartile Q3

57, 54, 54, 56, 57, 58, 58, 60, 61

Middle Value = 57

Q3 = 57

Five-number Summary (Age)

Minimum = 23

Q1 = 39

Median = 51

Q3 = 57

Maximum = 62

2. Fat Box Plot.

Step 1: Median (Q_2)

$$\frac{30.2 + 31.2}{2} = 30.7$$

$Q_2 =$

$Q_1 = 26.5$

$Q_3 = 34.1$

E) Step 1: Formulas.

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

$$\sum x = 836$$

$$\sum y = 518.6$$

$$\sum x^2 = 42498$$

$$\sum y^2 = 162368.59$$

$$\sum xy = 21763.2$$

$$r = \frac{18(21763.2) - 836(518.6)}{\sqrt{[18(42498) - 836^2][18(162368.59) - 518.6^2]}}$$

$$r = 0.92$$

As the value of Pearson Correlation is close to ± 1 . Age and percentage of fat are having Strong Positive Correlation.